

Vikraant Singh

(248) 878-4248 | singhvikraant06@gmail.com | www.linkedin.com/in/vikraantsingh

EDUCATION

University of Michigan

B.S. in Data Science

Coursework: Entry-level Programming, Introduction to Data Structures, Calculus II

Ann Arbor, MI

May 2028

EXPERIENCE

Data Scientist

Innovation for Impact

Technologies: *Python, Scikit-learn, Pandas, Matplotlib, Github Actions*

Jan. 2025 – PRESENT

Ann Arbor, MI

- Developed modular Python statistics engine to support csv file inputs, a list of 5+ basic statistics calculations catered to user specified target columns.
- Refactored codebase for statistics and clustering blocks, enabling an increase in data input compatibility for Optilogic developers, and facilitating seamless integration across diverse platforms.
- Applied scikit-learn to streamline batch processing, leading to a significant reduction in processing time by 60% during batch analyses of large datasets averaging over 100,000 entries.

Software Engineer Intern

Dorle Controls

Technologies: *Python, Github Actions, Winstar*

June 2024 – Aug. 2024

Farmington Hills, MI

- Built raw Python-based debugging tool to accurately translate 20+ hexadecimal messages into readable formats, streamlining internal testing and communications across vehicular functions.
- Designed marine digital dashboard using Winstar tech, enhancing user interaction and increasing customer satisfaction by 40% across 100+ marine vehicles.
- Gained hands-on expertise in decoding and validating EV component messages via hexadecimal translation processes.

PROJECTS

MyHindu

Technologies: *Javascript, CSS, React.js, React Native, Firebase, Expo, Github Actions*

June 2025 – PRESENT

- Developed comprehensive learning management system with 25+ educational modules sourced from academic theses and scholarly commentaries, implementing Firebase-powered user authentication and progress tracking to maintain 100% data persistence across sessions.
- Engineered dynamic Hindu calendar system calculating 25+ festivals and holidays using complex lunar-based algorithms spanning 4+ cosmic ages, reducing manual data entry by 95% and providing automated yearly festival calculations
- Built high-performance timeline visualization component handling 20+ historical events with crisp spring animations, improving user engagement through interactive cards with 300ms response times.

Guru.AI

Technologies: *Python, LangChain, OpenAI GPT-4, FAISS, PyPDF, Pydantic, Vector Embeddings, Github Actions*

July 2025 – PRESENT

- Developed an intelligent Hindu scripture chatbot using LangChain agents and OpenAI GPT-4, enabling users to query sacred texts (Bhagavad Gita, Upanishads, Vedas) with contextually accurate responses and verse citations.
- Implemented semantic search capabilities with FAISS vector database and OpenAI embeddings, processing 1000+ scripture chunks with 200-token overlap to ensure comprehensive knowledge retrieval and source attribution.
- Designed multi-tool agent architecture integrating scripture search, Google Search, and DuckDuckGo APIs with structured Pydantic response formatting, enabling comprehensive research capabilities and consistent JSON output for future API integration.

TECHNICAL SKILLS

Languages: Python, C++, Java, Javascript, CSS, Swift

Technologies: Customtkinter, Scikit-learn, Matplotlib, Datetime, Pandas, Smtplib, React.js, React Native, Firebase, Langchain